Vector Calculus and Complex Analysis Classwork (2025/4/21)

Name:

Student ID:

Show that the function $u(x,y)=3x^2y-y^3$ is harmonic, and find its harmonic conjugate $v(x, y)$. Express $f(z)=u+iv,\quad z=x+iy$ as an analytic function of z.

Solution

$$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=6y-6y=0$$

$$\frac{\partial v}{\partial y}=\frac{\partial u}{\partial x}=6xy \quad \Rightarrow \quad v=\int 6xydy=3xy^2+h(x)$$

$$\frac{\partial u}{\partial y}=-\frac{\partial v}{\partial x} \quad \Rightarrow \quad 3x^2-3y^2=-3y^2-h'(x)$$

$$h'(x)=-3x^2$$

$$h(x)=-x^3$$

$$v=3xy^2-x^3$$

$$f(z)=u+iv=3x^2y-y^3+i(3xy^2-x^3)=-iz^3$$